Skin, Hair, and Musculoskeletal Complications

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54.1 Complications Involving the Skin and Hair

54.1.1 Introduction

Nearly every recipient of an allo-HCT will at some stage develop complications involving the skin and hair. These complications can be grouped into drug-related toxicities and allergies, GVHD, infections, and malignant conditions.

54.1.2 Allergies, Drug-Related, and Other Toxicities

Drug-related toxicities are most often due to the conditioning regimen, antibiotics, or immunosuppressive (IS) agents. Presentation can vary broadly from localized erythema to epidermal necrolysis and Stevens-Johnson syndrome.

Diagnosis may be difficult because the morphological and chronological presentations of the lesions are generally nonspecific. Skin biopsies and histological examination can help improve diagnostic certainty (Paun et al. 2013).

Management requires discontinuation of suspected causative agents, topical treatment with healing ointments, prevention of secondary infections, and in severe cases (or when other organs are involved) systemic therapy with corticosteroids and antihistamines.

Extensive sun exposure including skin burns may increase the risk of skin GVHD, dryness, and subsequent infections. Regular examination of skin and skin-derived tissue and specialist consultation may enable early detection and better management (Majhail et al. 2012).

54.1.3 Graft-Versus-Host Disease

The skin is one of the most frequently affected organs in acute and chronic GVHD. Acute GVHD of the skin mainly affects the epidermis of the skin and adjacent oral, anal, and genital mucosa. Chronic GVHD may affect all layers of the skin including the epidermis, dermis, and subcutaneous tissue and may also manifest as skin dyspigmentation, sweat impairment, alopecia and thinning of scalp hair, hair loss in other areas (e.g., eye brows), and nail dystrophy. Chronic GVHD with sclerosis of the subcutaneous tissue including fasciae, joints, and the musculoskeletal system can severely impact patients' quality of life. Scleroderma lesions of the thorax or abdomen may impair breathing, lesions

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adjacent to joints may impair movement, and genital lesions may cause phimosis, vaginal scarring, and narrowing of the introitus that may cause dyspareunia and even complete obliteration of the vaginal tract.

In the early "inflammatory" phase of GVHD, patients often present with edema and discomfort which later progress to fibrosis and joint contractures. Regular survey of range of motion by patients and physicians may enable early detection of reversible lesions. Regular assessment (e.g., via questionnaire or oral interview) during routine clinical visits may encourage physicians and patients to address issues involving the genital tract, thereby enabling early detection and treatment.

General treatment and management of GVHD are discussed in Chaps. 43 and 44. In patients with chronic GVHD of the skin and subcutaneous tissues, treatment is best initiated in the early phase prior to development of most often irreversible fibrosis and contractures, which otherwise may require protracted immunosuppression and other measures, sometimes over several years. Physiotherapy including deep myofascial massage and stretching exercises is essential to restore or maintain range of motion. In patients with genital involvement, topical treatment with immunosuppressive agents and hormones and use of vaginal dilators should be initiated early to prevent or reduce the degree of irreversible fibrosis and avoid the need for surgical intervention.

54.1.4 Infectious Complications Involving the Skin

Infection-associated skin lesions are often due to viruses. Unexplained fever and rash are more frequent in patients with HHV6 viremia compared to controls (Betts et al. 2011). Due to the lack of effective prophylaxis, HHV6-related complications occur during early transplant phases, especially in PT-Cy setting, while shingles (varicella zoster) are mostly seen beyond 6 months after transplant and mainly after discontinuation of prophylactic acyclovir (Noviello et al. 2023). HSV1-related complications are more frequent in early weeks and also in patients under continuous immunosuppression for GVHD and may occur despite prophylactic aciclovir. Other infectious conditions of the skin include fungal infections (mainly due to dermatophytes and, less frequently, *Aspergillus* or *Mucor* species) and bacterial infections. Management of infections mostly consists of systemic antiviral, antifungal, or antibiotic treatment.

54.1.5 Malignant Complications of the Skin

Post-transplant malignant conditions of the skin include skin cancer (basal cell carcinoma, squamous cell carcinoma, and melanoma) and relapse of underlying malignant disease. BCC and SCC are much more common and have a better prognosis than melanoma. The incidence of melanoma has been reported to be higher after allo-HCT with standardized incidence ratios ranging from 1.4 to 8.3 (Inamoto et al. 2015). Secondary cancers, risk factors, and management are discussed in detail in Chap. 47. Overall, the survival for these patients is comparable to those of patients with the same de novo cancer (Tichelli et al. 2019).

Patients should be counseled to perform self-examination of the skin and adjacent mucosa, use adequate sun protection, and avoid excessive sun exposure. Country-specific general population recommendations for screening for cancer should be adapted and modified taking increased risk of HCT survivors into consideration.

54.2 Musculoskeletal Complications

54.2.1 Introduction

Complications involving the muscles include myopathies, myositis, and cramps. Musculoskeletal complications are reported in 35% of long-term survivors 10 years after allogeneic transplantation (Syrjala et al. 2005).

54.2.2 Myopathy

The most frequent causes of myopathy early after transplant are corticosteroid therapy and inactivity. Patients report muscle weakness with no pain, and laboratory investigations show normal creatine kinase. The proximal lower limb muscles, particularly the quadriceps muscles, are most severely affected. The main risk factors include increasing dose and duration of corticosteroid therapy, older patient age, and the extent and duration of inactivity, particularly when intensive care is required. Patients should receive physiotherapy as soon as corticosteroid therapy is initiated and be advised to exercise on their own (Mohammed et al. 2019). Systemic corticosteroids should be tapered or avoided when possible.

54.2.3 Myasthenia Gravis

Though rare (<1%), it has been reported after allogeneic hematopoietic cell transplantation, mostly in the context of chronic GVHD. Patients present with fatigable weakness during or after tapering of immunosuppression (Grauer et al. 2010). Diagnosis includes detection of antibodies against acetylcholine in blood. Treatment consists of cholinesterase inhibitors and IS therapies for chronic GVHD.

54.2.4 Muscle Cramps

Muscle cramps are painful and often visible contractions lasting up to 30 min. Though rarely reported, they appear to be frequent in patients with chronic GVHD (Filipovich et al. 2005); an association with chronic IS may also be possible. Magnesium deficiency and side effects of medications (e.g., ganciclovir, valganciclovir) should always be ruled out. If magnesium replacement and discontinuation of suspected causative drugs do not bring relief, treatment with, for example, quinine or antiepileptic drugs may be considered.

54.2.5 Myositis

Myositis has been reported in up to 3% of patients after allo-HCT. Though frequently associated with other symptoms of chronic GVHD, it can also be the sole manifestation of GVHD (Openshaw et al. 2009). Patients often present with pressure-sensitive muscle pain and increased blood creatine kinase. Management is within GVHD treatment (Couriel et al. 2002).

54.2.6 Complications Involving the Bones and Joints

The most frequent complications involving the bones and joints are chronic GVHD, avascular osteonecrosis, and bone loss (osteopenia/osteoporosis). Chronic GVHD of the joints is discussed in Chap. 44.

54.2.6.1 Osteoporosis/Osteopenia

Osteopenia (defined as a T-score between −1 and −2.5) and osteoporosis (defined as a T-score less than −2.5) have been reported in about 25–50% of patients after allo-HCT and up to 60% in patients with severe chronic GVHD (Pirsl et al. 2016). Risk factors include protracted IS, older patient age, higher cumulative corticosteroid dose (Schulte and Beelen 2004; Savani et al. 2007; Stern et al. 2001; Yao et al. 2008; Petropoulou et al. 2010; Abou-Mourad et al. 2010), lower body weight, malnutrition, physical inactivity, female gender, higher average NIH organ score, as well as higher platelet counts in patients with severe chronic GVHD (Pirsl et al. 2016).

In accordance with these risk factors, decrease in bone mineral density occurs most rapidly within the first year after transplant. If osteopenia or osteoporosis is diagnosed, endocrine causes like hyperthyroidism, hyperparathyroidism, and hypogonadism need to be ruled out. Screening using dual energy X-ray absorptiometry (DEXA) is recommended 1 year after transplant and repeat measurements in patients with recognized defects (Majhail et al. 2012).

Measures to prevent bone loss include vitamin D supplementation in regions with high prevalence of vitamin D deficiency, adequate calcium intake preferable through diet, and regular weight-bearing physical exercise. Beyond the above preventive measures, specific treatment is recommended for patients with severe osteopenia or osteoporosis (Bhatia et al., 2017). Hormone replacement therapy should be considered in patients with hypogonadism. Reduction in the risk of fractures has been demonstrated for several agents and agent combinations (Barrionuevo et al. 2019). Patients should also be counseled to modify negative lifestyle factors (e.g., cease smoking) and take measures to prevent falls (e.g., physical exercise including balance training, correct visual disorders).

54.2.6.2 Avascular Necrosis

Avascular necrosis (AVN) has been reported in up to 19% of adult patients and up to 29% of patients younger than 20 years (Torii et al. 2001; Patel et al. 2008). Risk factors include GVHD, steroid therapy, microvascular changes due to GVHD and/or its therapy, younger age at transplant, and TBI (Socié et al. 1997; French et al. 2008; Patel et al. 2008; Campbell et al. 2009; Jagasia et al. 2010). Patients usually present with joint pain, restricted to one or two affected joints. Though most joints can be affected, the hips are by far most frequently involved, bilateral in the majority of cases.

Screening for AVN is not recommended; however, high index of suspicion and prompt MRI are necessary in early symptomatic patients with risk factors, to enable early detection and intervention (Bhatia et al. 2017).

Pain relief and maintenance or restoration of patient mobility are the main aims or treatment. Discontinuation of corticosteroid and other IS therapies should be considered where possible. Drug therapy is limited to pain relief. The role of nonsurgical causative therapies, such as bisphosphonates, statins, and prostacyclin analogues, is still unclear. Pressure relief by means of surgical core decompression may relieve pain and slow down progression in early stages, whereby additional autologous bone marrow grafting further improves long-term outcome (Hernigou et al. 2018). In patients with late-stage disease with femoral head collapse, joint-preserving strategies are not effective, and total hip arthroplasty is the recommended long-term treatment.

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